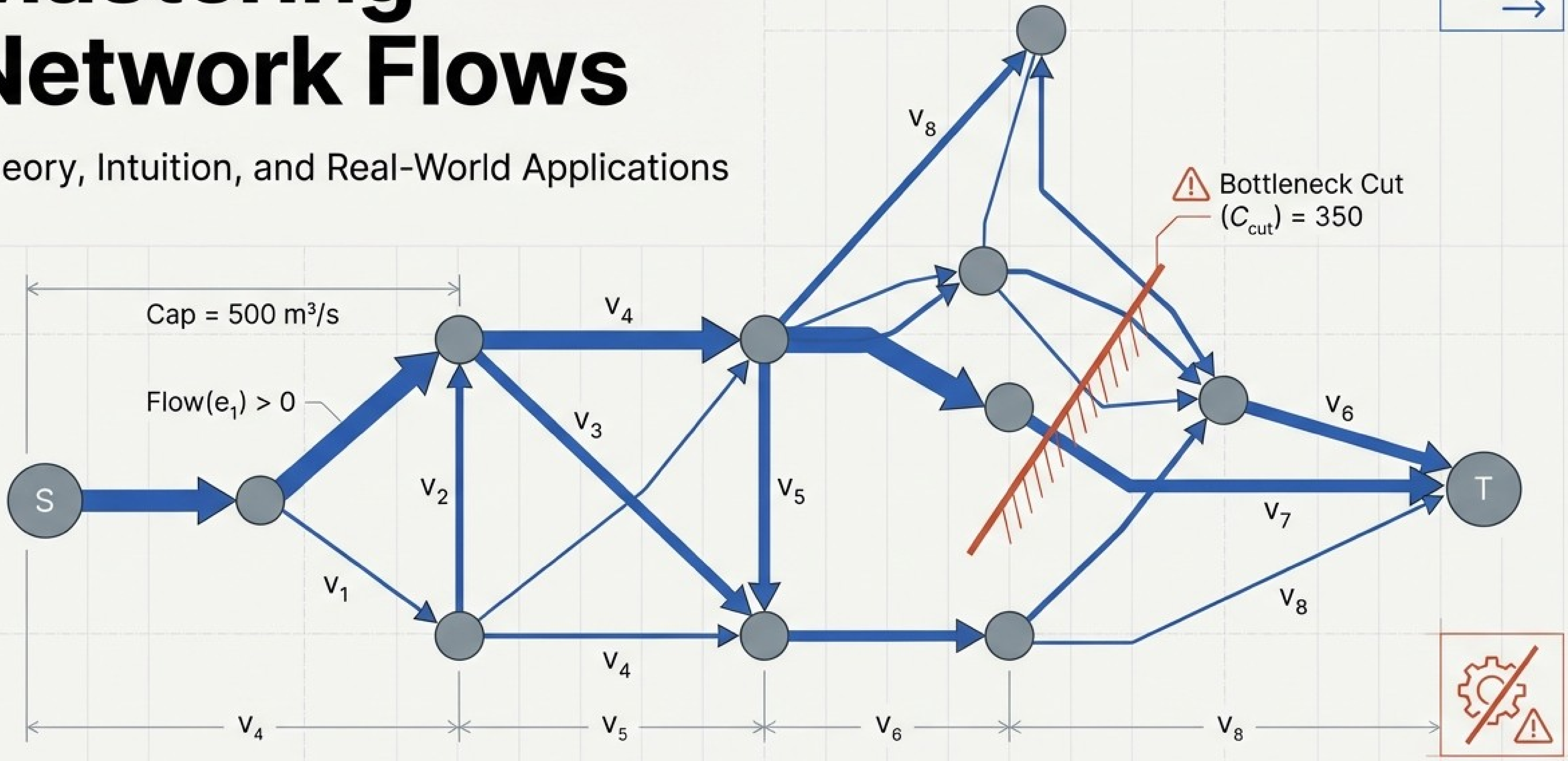
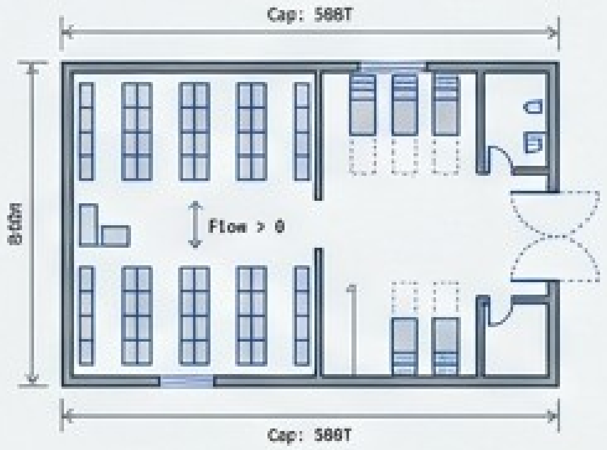
Mastering Network Flows

Theory, Intuition, and Real-World Applications



Mathematical engines powering modern infrastructure

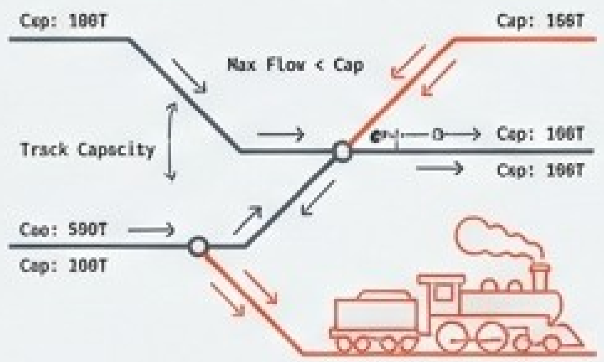


Supply Chain Logistics

The Node
Regional warehouses & retail stores.

The Edge
Transportation routes & trucking limits.

Ultimate Goal
Find minimum-cost flow to route goods optimally, minimizing fuel/tolls and saving thousands daily.

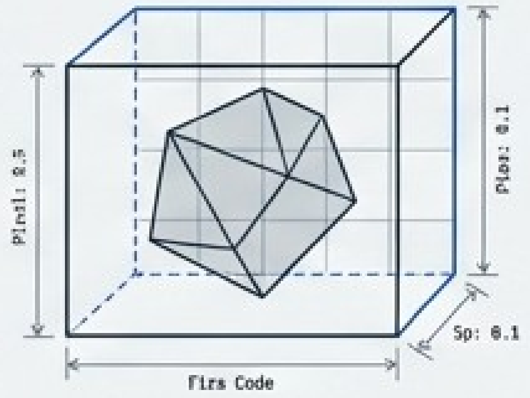


Military & Railway Traffic

The Node
Rail junctions.

The Edge
Train track tonnage limits.

Ultimate Goal
Pinpoint the central bottleneck of an enemy transportation network to understand maximum supply rates.



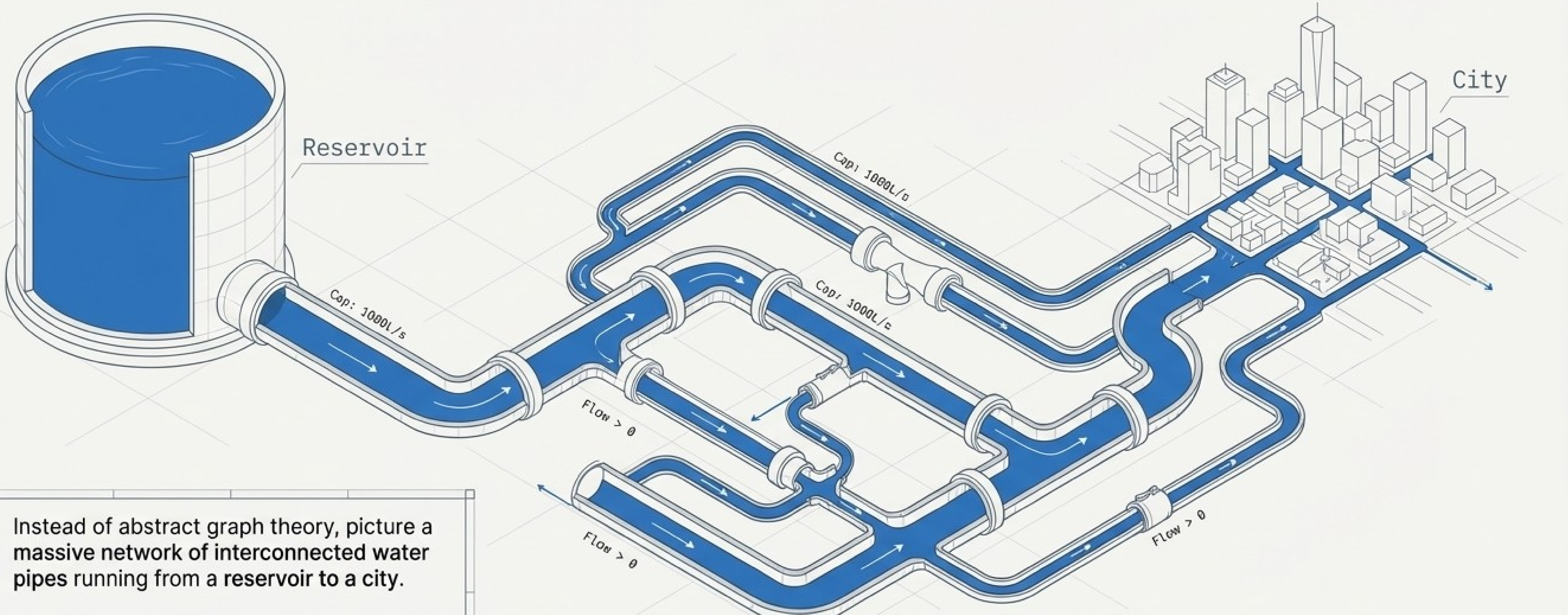
Computer Vision

The Node
Individual image pixels.

The Edge
Pixel similarities and separation penalties.

Ultimate Goal
Apply minimum cut algorithms to sever the graph at weakest links, separating foreground / background.

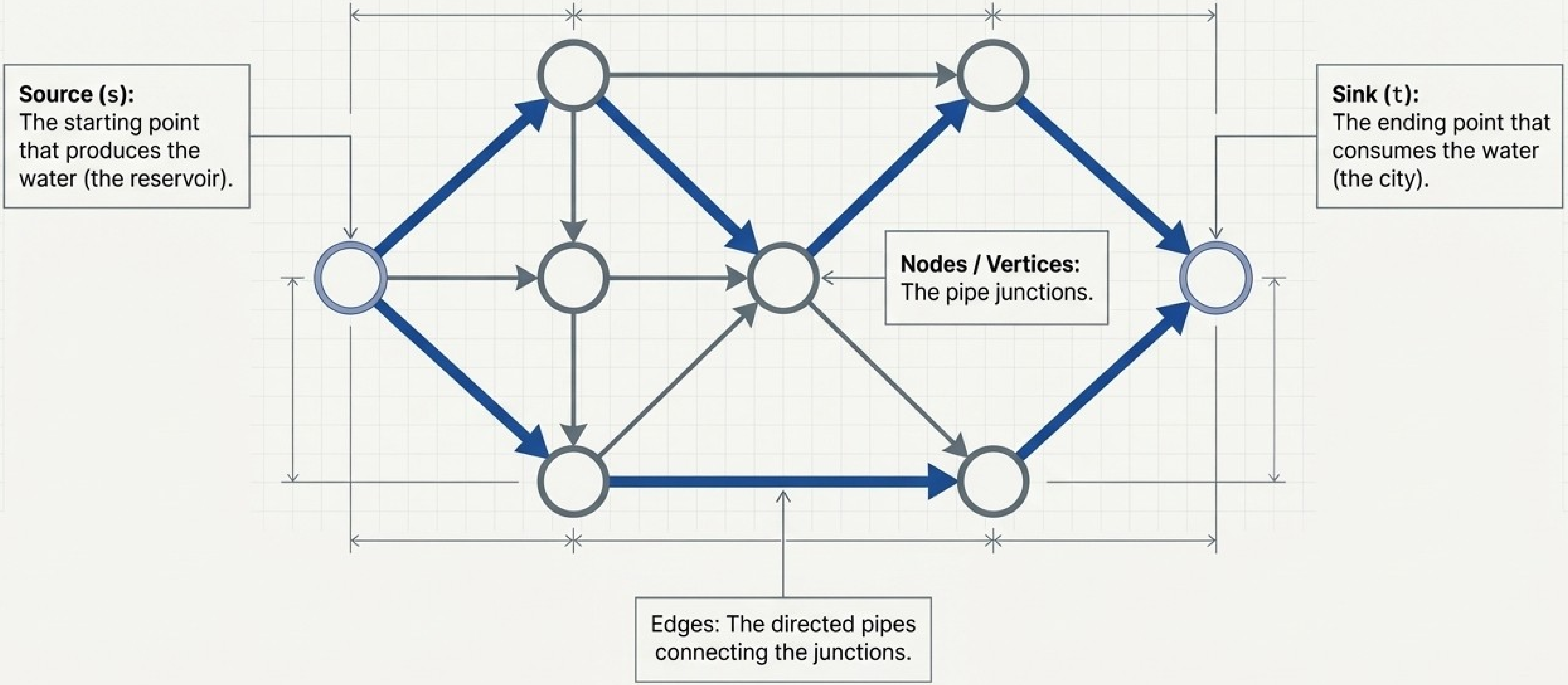
The universal mental model is physical fluid dynamics



Instead of abstract graph theory, picture a **massive network of interconnected water pipes** running from a **reservoir to a city**.

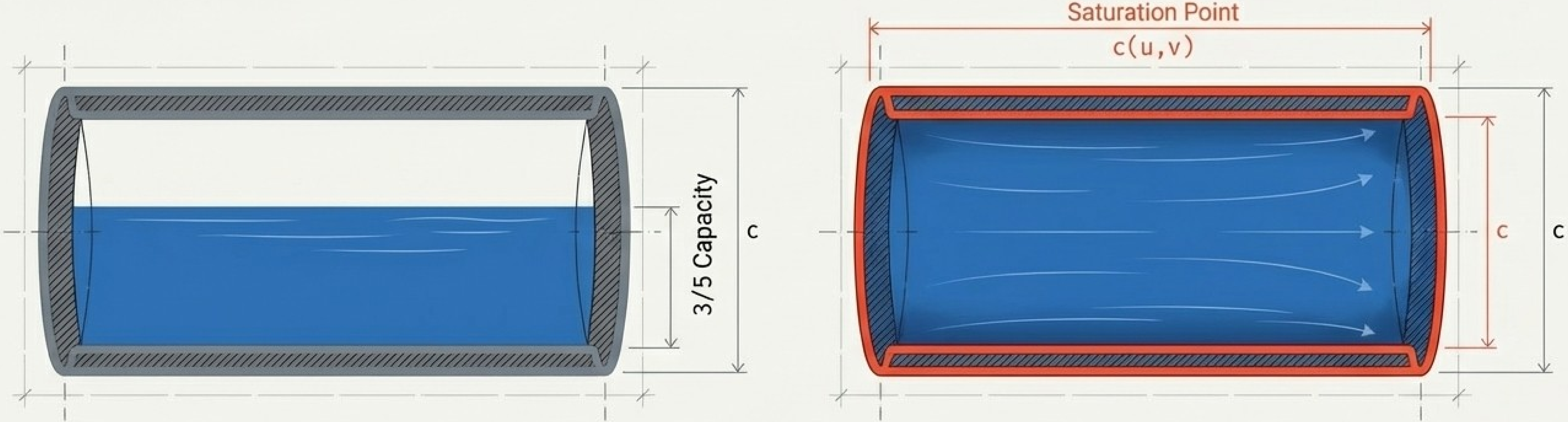
Every mathematical rule of network flows is simply a rule of physical water and pipes.

Anatomy of a Flow Network



The absolute limits of the pipe

Every pipe has a specific diameter, dictating the maximum volume of water it can carry. The flow cannot exceed this given capacity: $0 \le f(u,v) \le c(u,v)$



Partially Filled

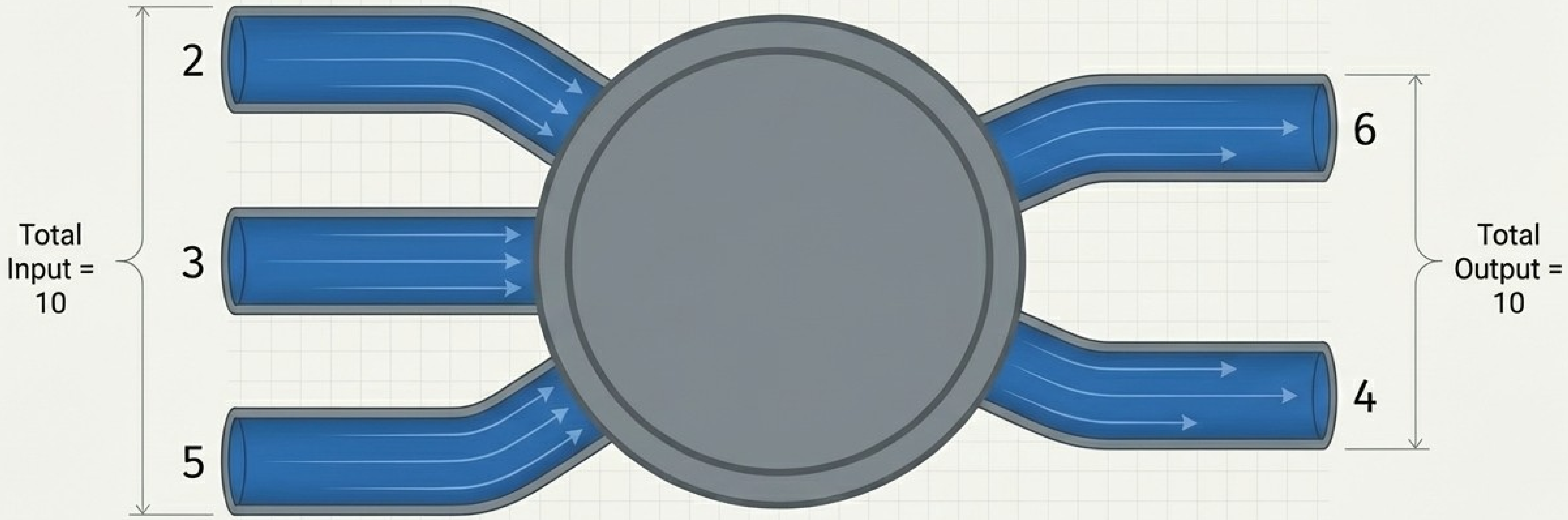
3/5

Completely Saturated

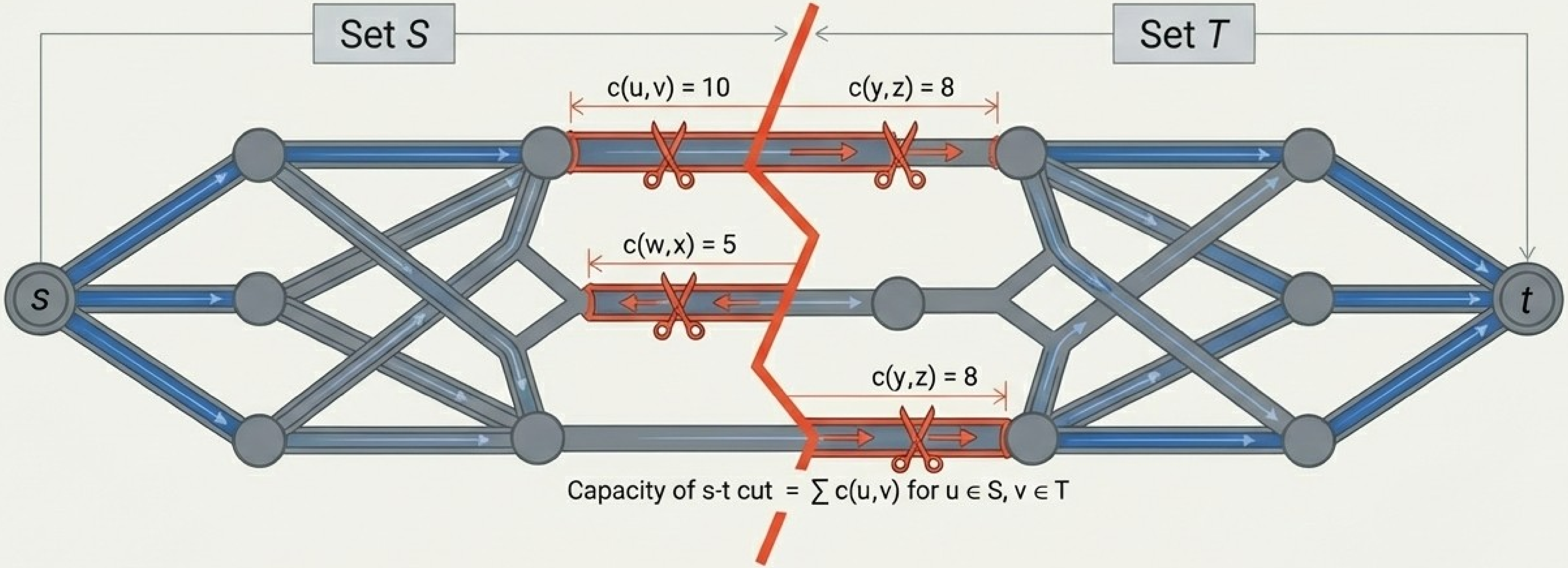
5/5

The Flow Conservation Law

Water cannot magically multiply, nor can it leak out into nothingness. For every intermediate node, the net flow entering the node is exactly zero.



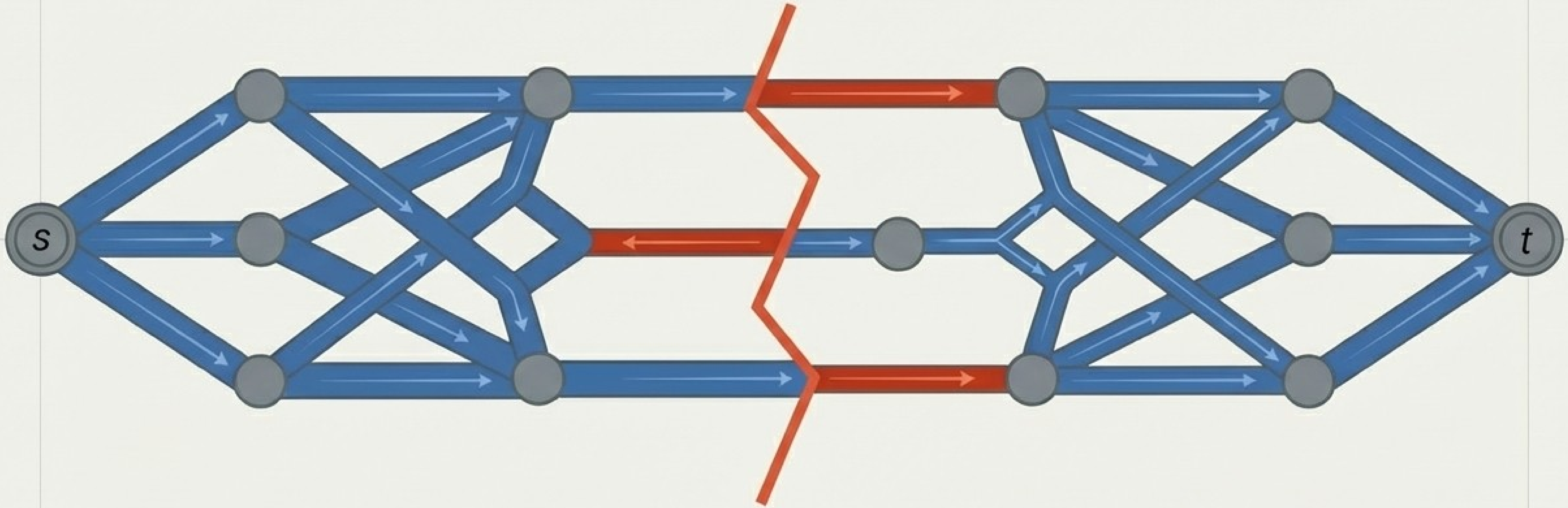
The core physical question: The Enemy's Cut



If an enemy wanted to completely cut off the city's water supply from the reservoir, what is the absolute cheapest way they could do it? The capacity of this s-t cut is the sum of the capacities of the pipes crossing the slice.

The Max-Flow Min-Cut Theorem

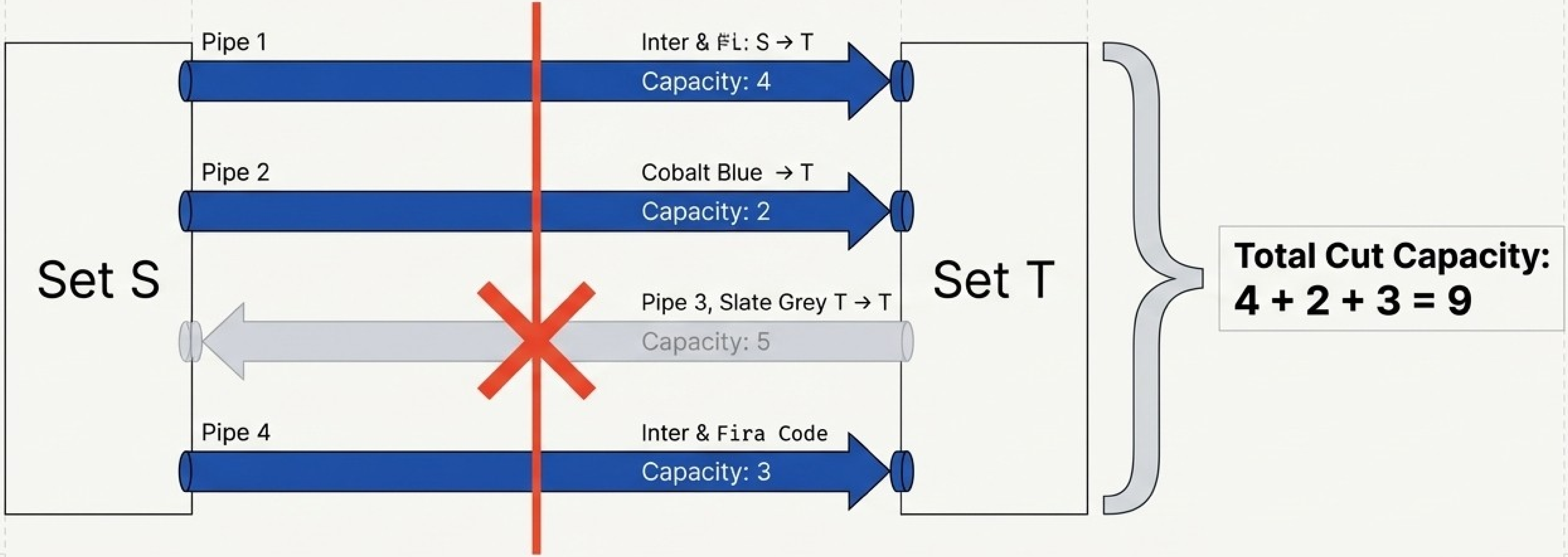
The maximum value of an s-t flow is exactly equal to the minimum capacity over all s-t cuts.



The profound beauty of duality: The entire network is only as strong as its weakest link. Solving for the maximum flow mathematically simultaneously solves for the network's critical bottleneck.

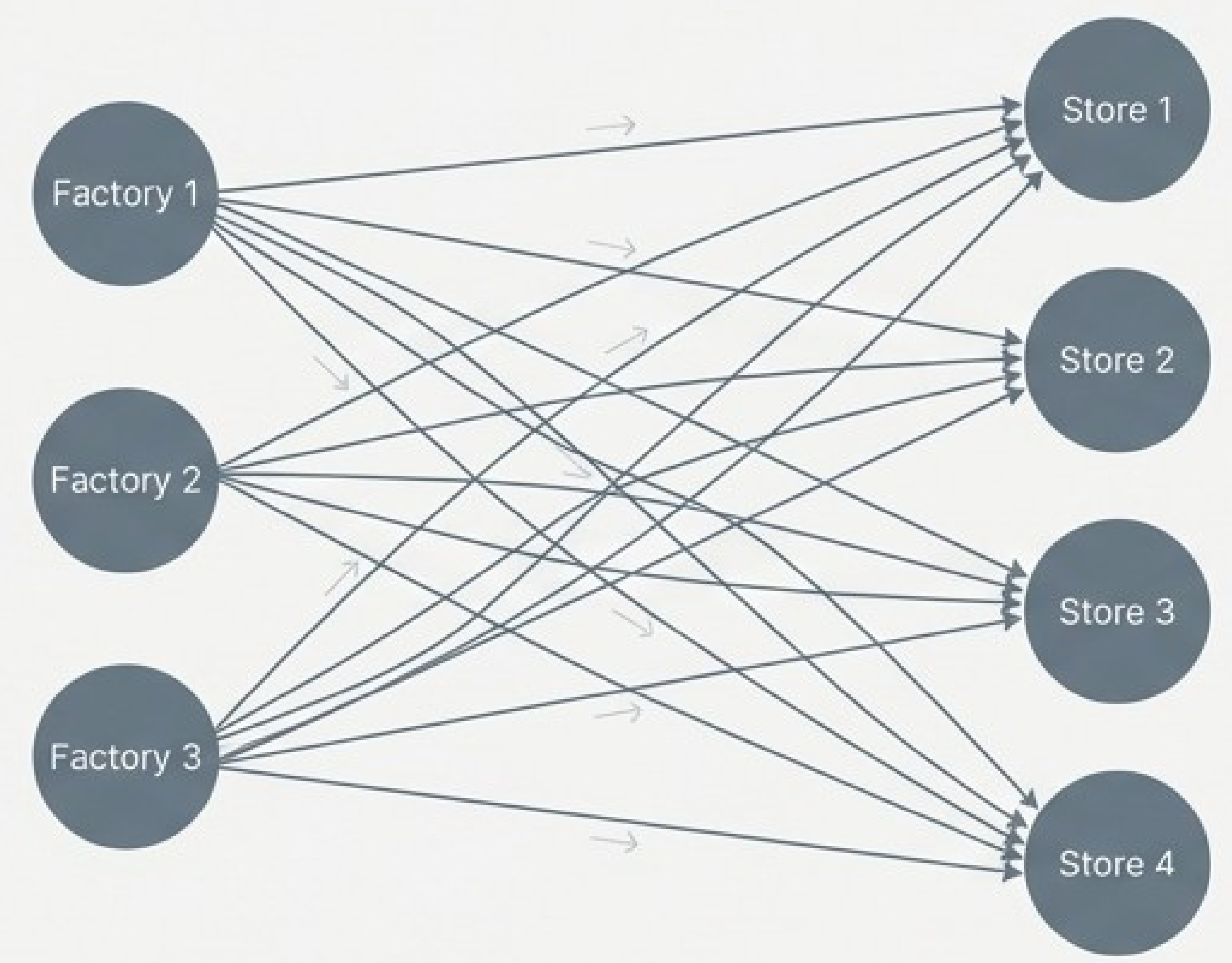
Anatomy of a Cut: Directionality matters

When calculating the capacity of a cut, do we sum up all the edges? No. We only sum the capacities of edges directed from the source set to the sink set. Backward edges are ignored.

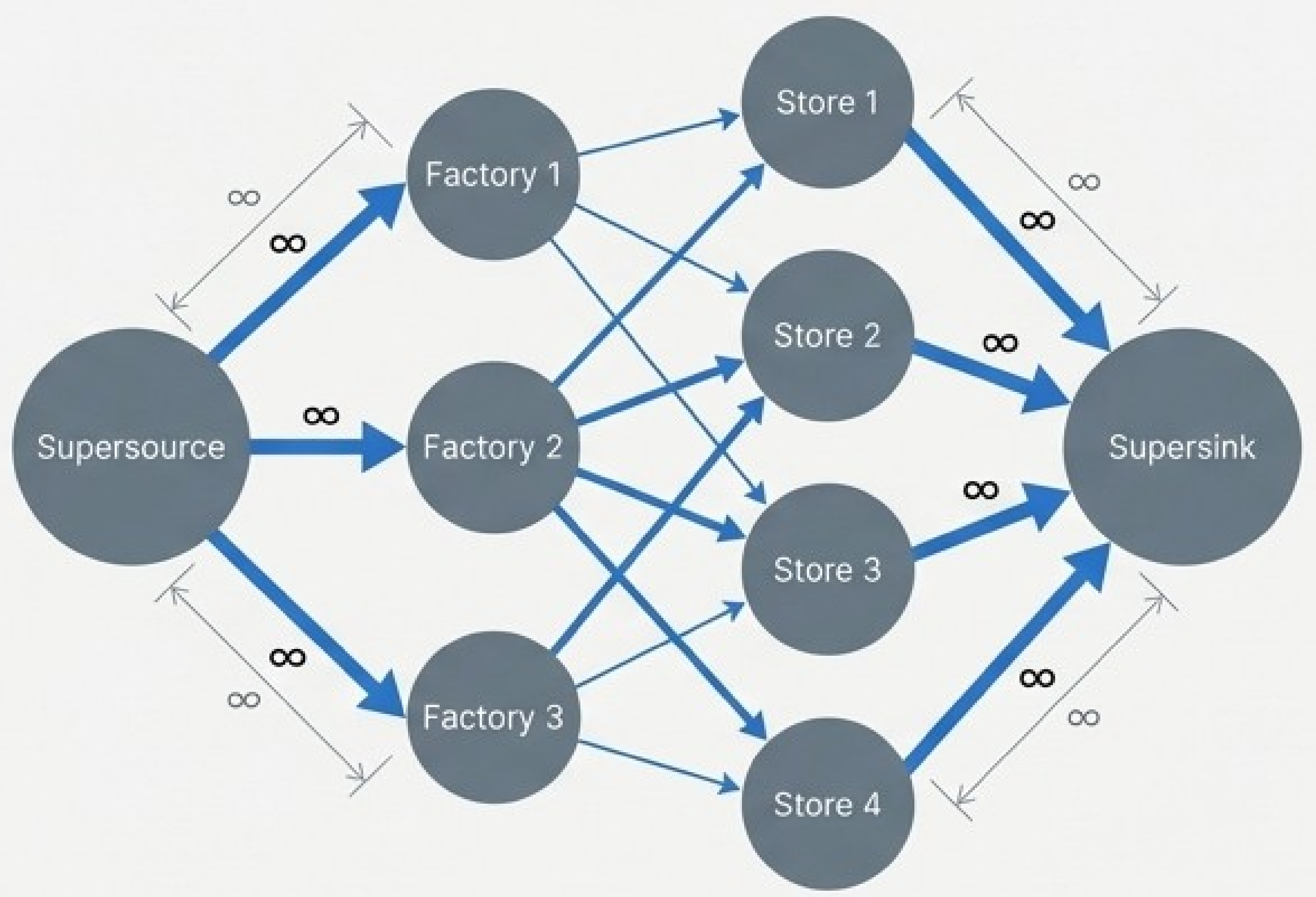


Scaling up: Multiple sources and multiple sinks

The Problem: Complex Network



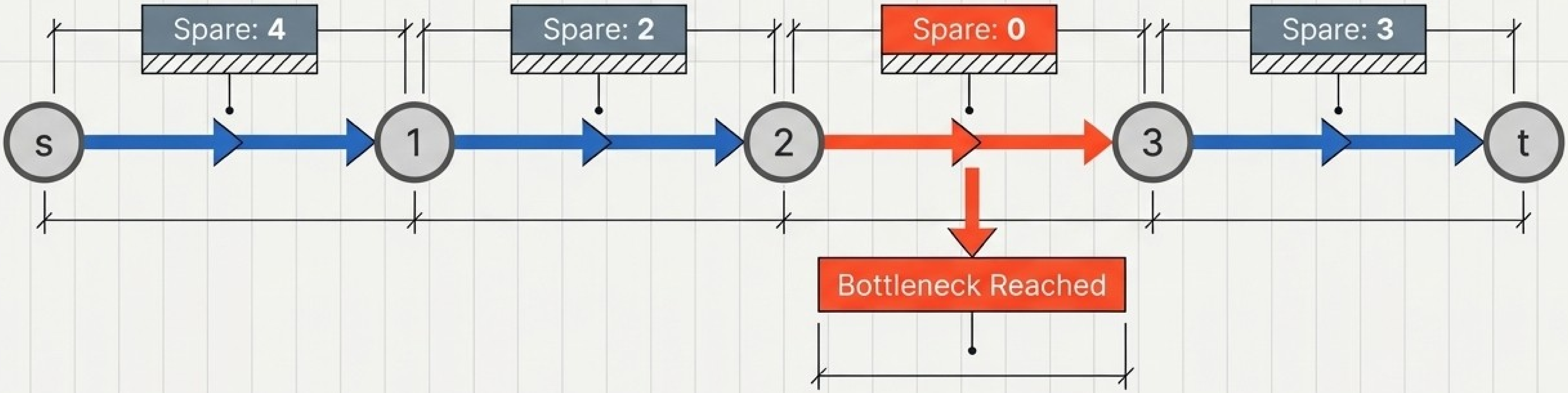
The Solution: Supersource & Supersink



To model real-world scenarios with multiple factories and stores, we introduce a supersource and supersink with infinite capacity edges. This elegantly collapses complex infrastructure back into a standard single-source, single-sink network.

The stopping condition: Augmenting Paths

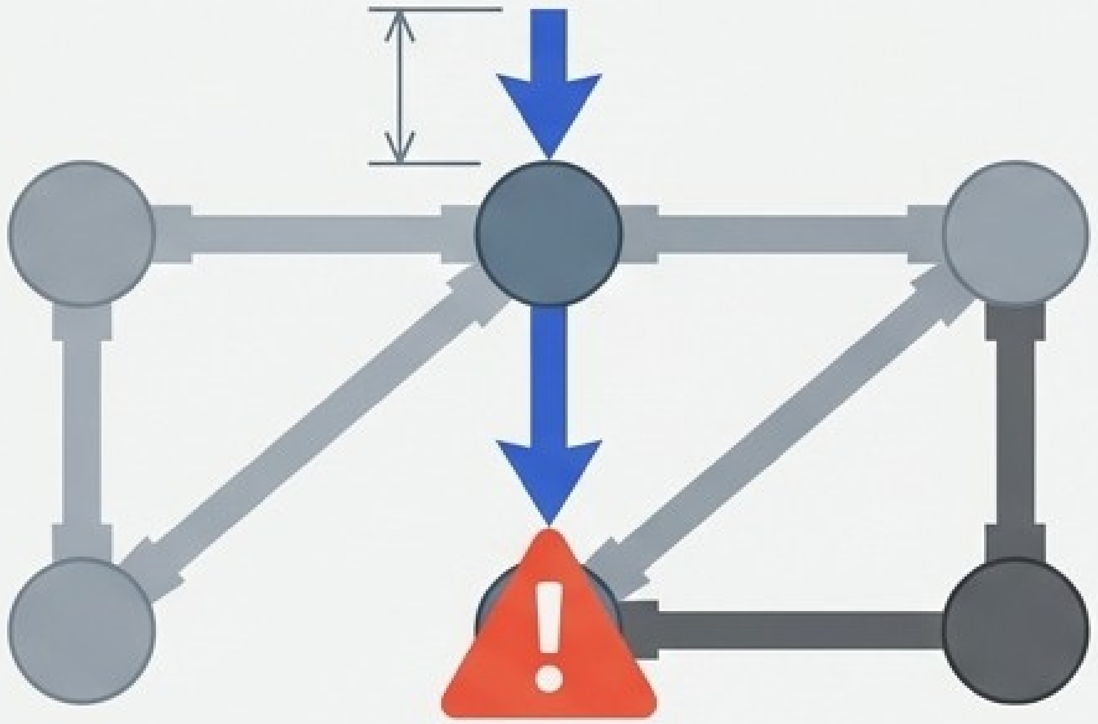
How does the algorithm “know” it has found maximum flow? It continually searches for an augmenting path: an available route with spare capacity. Once spare capacity hits zero and no such path exists, true maximum flow is achieved.



The Undo Mechanism: Residual Graphs

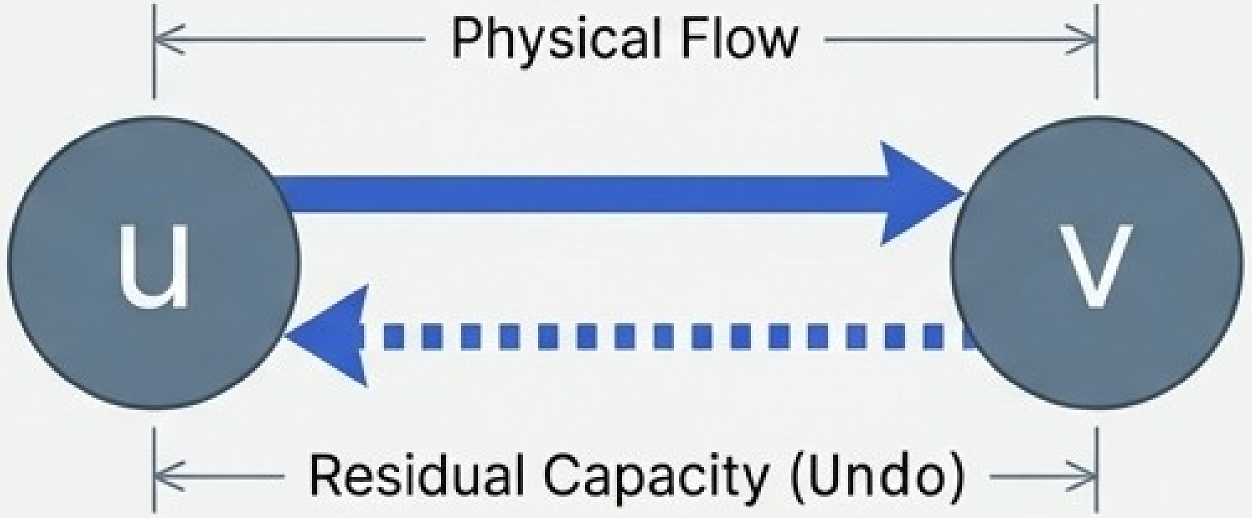
A purely greedy approach might make a bad initial choice that blocks the optimal path. Residual graphs feature “backward edges” that allow the algorithm to dynamically undo bad decisions.

The Wrong Way: Greedy Push



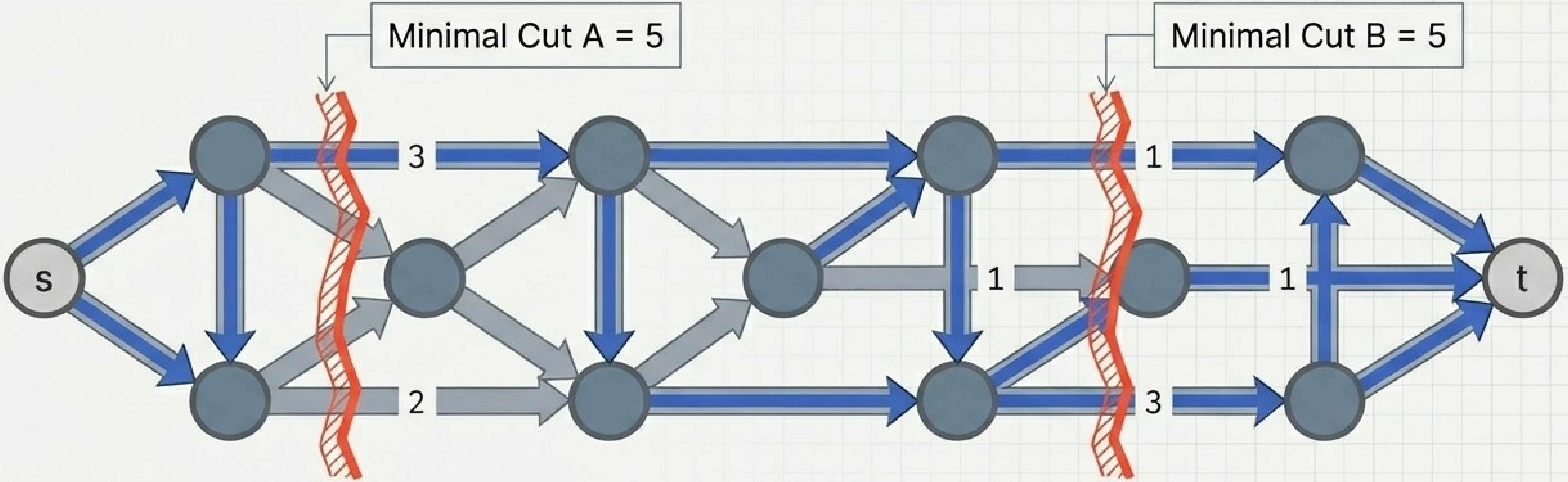
Gets Trapped

The Right Way: Dynamic Residuals



System Ties: Dealing with multiple bottlenecks

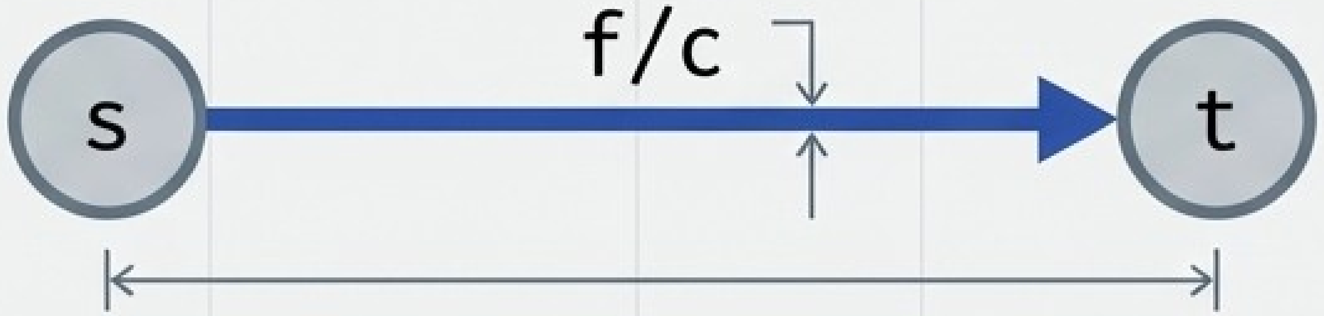
What if there is a tie? A network can have multiple minimal cuts sharing the exact same capacity value. The theorem guarantees the maximum flow will max out at this exact minimum value.



Any of these minimal cuts represents a valid, fully saturated system bottleneck.

Network Flows: The Unified Cheat Sheet

The Anatomy



The Core Laws

1. Capacity Constraint: $0 \leq f(u,v) \leq c(u,v)$
2. Flow Conservation: Net flow entering any intermediate node = 0.

The Theorem

Max Flow = Min Cut.

The maximum amount of flow mathematically dictates the exact minimum physical bottleneck of the system.

Algorithmic Mechanics

- - Only sum S → T edges for cuts.
- - Use infinite-capacity Supersource/Supersinks for multiple origins.
- - Residual graphs enable backward "undo" flows.
- - Zero augmenting paths = maximum flow achieved.